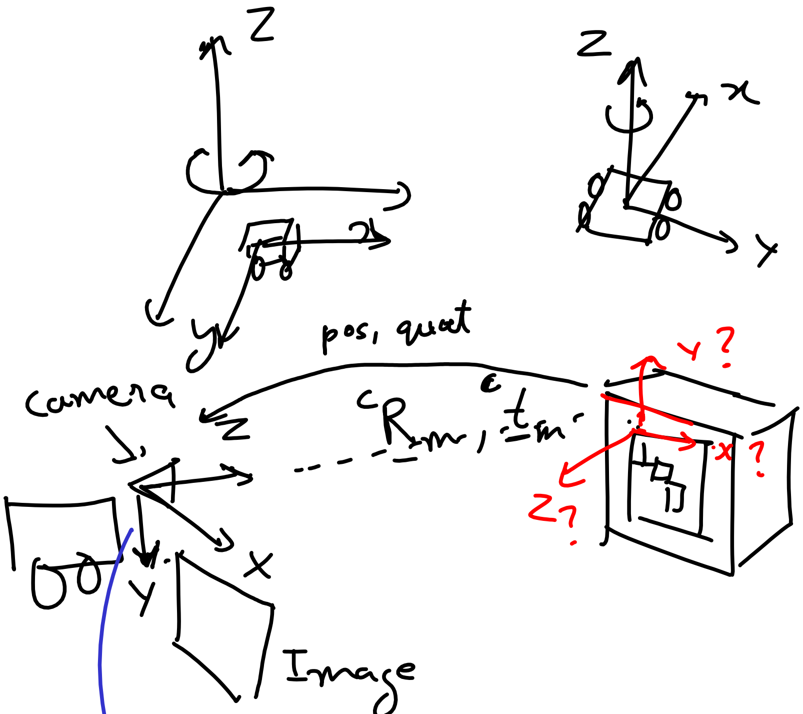
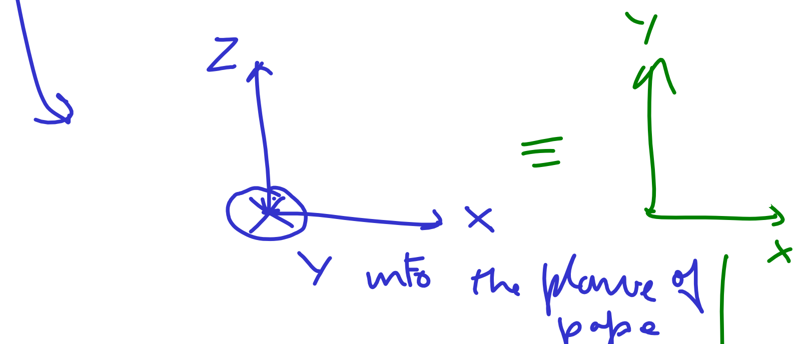
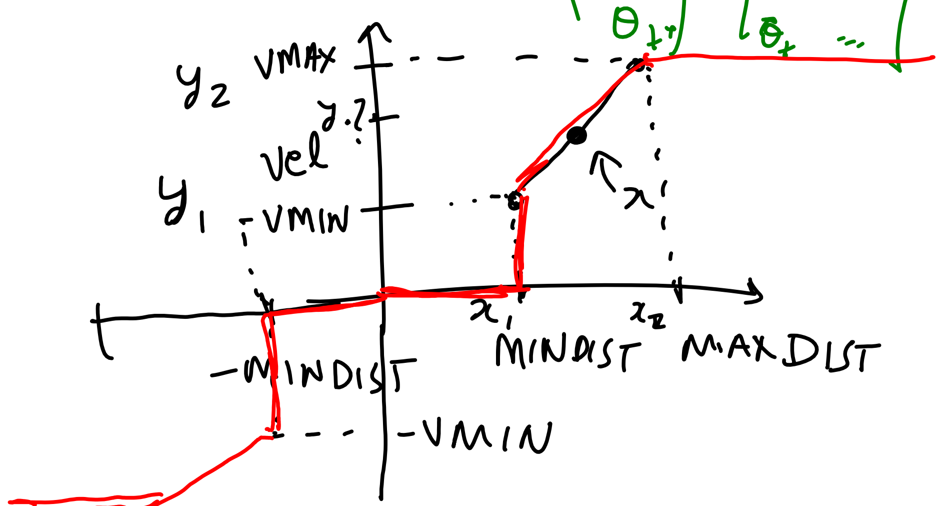




$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{pmatrix} = \begin{pmatrix} x_t + V_t \cos\theta_t \\ y_t + \\ \theta_t \ \cdots \end{pmatrix}$$

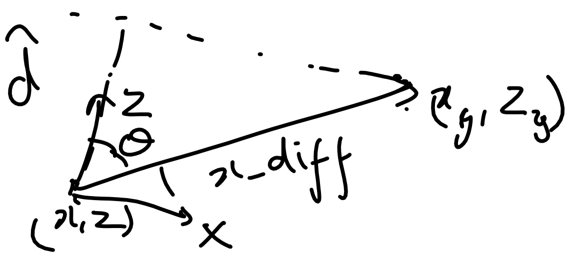


$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y - y_1 = m(x - x_1)$$

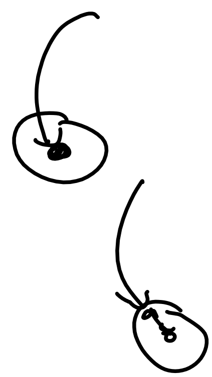
$$\Rightarrow y = y_1 + m(x - x_1)$$

z MINUS



$$\underbrace{\hat{d}^T \underline{x}\text{-diff}}_{z}$$

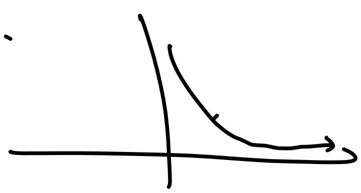
① P-Only controller



② PI controller

$$a_t = \underbrace{k_I \int_0^t e(t)\,dt}_{\text{accumulate small errors and correct for them}} + k_p e(t)$$

accumulate
small errors
and correct for them



③ PID controller

$$a_t = \underbrace{k_d \frac{d}{dt} e(t)}_{\text{respond faster}} + \text{PI terms}$$

respond faster
(avoid oscillations) to change in direction of error